

# WU SI

Ph.D., The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong

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## WORK EXPERIENCE

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**The Hong Kong University of Science and Technology**, Postdoctoral Fellow **2025 - now**

Supervisor: Prof. WANG Zhe

Research interests: Energy Efficient Data Center; Predictive Maintenance.

## EDUCATION

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**The Hong Kong University of Science and Technology**, Ph.D. in Civil Engineering **2022 - 2025**

Thesis: *Enhancing HVAC System Efficiency: Development of Virtual Testbed and Evaluation of Equipment Performance*

Supervisors: Prof. WANG Zhe and Prof. CHEN Guanghao

\*HKUST RedBird Academic Excellence Award for Continuing PhD Students (2025)

**Shanghai Jiao Tong University**, M.Sc. in Power Engineering and Thermophysics **2019 - 2022**

Thesis: *Numerical Investigations on a Turbofan Afterburner Fuel Pump*

Supervisors: Prof. OUYANG Hua and Prof. WU Yadong

**Dalian University of Technology**, B.E. in Process Equipment and Control Engineering **2015 - 2019**

## PUBLICATIONS

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1. **Wu, S.**, Yang, P., Li, D., Chen, G., & Wang, Z.\* (2025). Towards predictive maintenance: A performance evaluation framework for cooling towers in HVAC systems. *Building and Environment*, 284, 113443.  
<https://doi.org/10.1016/j.buildenv.2025.113443>
  2. **Wu, S.**, Zheng, W., Wang, Z.\*, Chen, G.\*, Yang, P., Yue, S., ... Wu, Y. (2025). *AlphaDataCenterCooling*: A virtual testbed for evaluating operational strategies in data center cooling plants. *Applied Energy*, 380, 125100.  
<https://doi.org/10.1016/j.apenergy.2024.125100>
  3. **Wu, S.**, Yang, P., Chen, G., & Wang, Z.\* (2025). Evaluating seasonal chiller performance using operational data. *Applied Energy*, 377, 124377.  
<https://doi.org/10.1016/j.apenergy.2024.124377>
  4. **Wu, S.**, Wu, Y.\*, Tian, J., & Ouyang, H. (2022). On the cavitation-induced collapse erosion of a turbofan fuel pump. *Engineering Applications of Computational Fluid Mechanics*, 16(1), 1048–1063.  
<https://doi.org/10.1080/19942060.2022.2067243>
  5. Yang, Z., Ming, L., **Wu, S.**, Wu, Y., Tian, J., & Ouyang, H. (2022, June). On the mode characteristics of rotating instability with different tip clearances. In *Turbo Expo: Power for Land, Sea, and Air* (Vol. 86120, p. V10DT37A013). American Society of Mechanical Engineers.  
<https://doi.org/10.1115/GT2022-82072>

## RESEARCH EXPERIENCE

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**Model Predictive Control for Energy Efficient Data Center** **Feb. 2023 - June 2024**

*School of Engineering, HKUST*

Developed an open-sourced virtual testbed *AlphaDataCenterCooling* for data center cooling plant control strategy optimization.

<https://github.com/wfzheng/AlphaDataCenterCooling>

Proposed a method for benchmarking chiller performance using operational data without shutdown measurements.

**On the Cavitation-Induced Collapse Erosion of a Turbofan Fuel Pump** **Oct. 2020 - Aug. 2022**

*School of Mechanical Engineering, SJTU*

High-speed multi-phase flow field analysis based on computational fluid dynamics (CFD) simulations.

Optimized design of a turbofan fuel pump based on fluid-structure coupling simulations.

SERVICE

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|--------------------|------------------------------------------------------------------------------------|
| Teaching Assistant | Fundamental of Green Buildings<br>Energy System Modelling for Buildings and Cities |
| Journal Reviewer   | Building Simulation<br>Energy and Buildings                                        |